

ARTIFICIAL INTELLIGENCE (CSA1717)

ASSESSMENT – 2

PROBLEMS

Problem 1: Doctor Shift Scheduling using Backtracking Search

A hospital needs to assign three doctors (D1, D2, and D3) to three different shifts: Morning, Afternoon, and Night. The schedule must satisfy the following constraints: Doctor D1 cannot work the Night shift, Doctor D2 must be assigned before Doctor D3 according to the order Morning < Afternoon < Night, only one doctor can be assigned to each shift, and Doctor D3 cannot be assigned to the Morning shift. Every doctor must be assigned exactly one shift. Apply the Backtracking Search Algorithm to determine a valid schedule while showing the assignment order, constraint checking, intermediate steps, and the final shift allocation.

Problem 2: Robot Grid Navigation using Breadth First Search (BFS)

A robot is placed in a 5×5 grid environment and must move from the Start (S) position to the Goal (G) position while avoiding obstacles. The robot can move only in four directions: Up, Down, Left, and Right, and each movement has a cost of 1. The robot cannot pass through blocked cells and must remain within the boundaries of the grid. Apply the Breadth First Search (BFS) algorithm to determine the shortest path from the start node to the goal node. Show the queue (frontier) updates after each node expansion, calculate the Manhattan Distance heuristic, and display the final path along with the total path cost.

Problem 3: Autonomous Rescue Robot using Uniform Cost Search

An autonomous rescue robot operates in a disaster-affected environment where obstacles, damaged roads, and unknown paths may exist. The robot must safely navigate the environment, locate victims, and reach them by selecting the least-cost path. The robot should behave as a rational intelligent agent by continuously observing its surroundings, making appropriate decisions, and responding to changes in the environment. Identify the suitable Intelligent Agent for this scenario, explain the environment characteristics and the rationality of the agent, and describe how Uniform Cost Search (UCS) or an Online Search Agent can be used to efficiently determine the safest and least-cost path for successful rescue operations.

